

AI & Machine Learning - Revision Notes

Personal Study Guide

Started: March 14, 2026 | Instructor: AI Tutor (Claude)

1. What is Intelligence?

Ability to learn from experience, understand new situations, make decisions, and adapt behavior.

- Learn + Understand + Decide + Adapt = Intelligence

Dark Clouds: See clouds -> remember rain -> carry umbrella. Video Game: first play terrible, slowly learn patterns.

2. What is AI?

Teaching machines to do things that require human intelligence. AI = brain (software), Robot = body (hardware).

- Most AI is software, NOT physical robots
- Examples: Instagram recommendations, Gmail spam filter, Google Maps

3. Types of AI

Type	What	Exists?
Narrow AI	One thing well	YES (all current AI)
General AI	Everything human can	Not yet
Super AI	Better than all humans	Sci-fi

- Even ChatGPT/Claude = advanced Narrow AI

4. Traditional Programming vs Machine Learning

Traditional: Rules + Data -> Answers (human writes logic)

ML: Data + Answers -> Rules (computer learns logic)

5. AI Hierarchy: AI -> ML -> DL -> LLMs

Nested like Russian dolls. DL = many layers. "Deep" = layers finding progressively complex patterns.

Term	Definition
Neural Network	Layers that find patterns
Weight	Importance number
Training	guess -> check -> adjust -> repeat
Loss	How wrong the guess was
Gradient Descent	Method to find which weights to fix
LLM	Deep Learning for language

6. Supervised Learning

Learning WITH labels. Data + correct answers -> model learns patterns.

Classification = answer is a **WORD** (spam, cat, S/M/L)

Regression = answer is a **NUMBER** (Rs 350K, 87 marks)

Classification Models

Model	How	Analogy
Logistic Regression	Line between groups	Fence
Decision Tree	Yes/No flowchart	20 Questions
Random Forest	100+ trees vote	100 doctors
SVM	Best boundary max gap	Wall with space
KNN	K closest neighbors	Rich neighbors

Regression Models

Model	How	Analogy
Linear Regression	Straight line	Size vs price
Polynomial	Curved line	Life not always straight
Random Forest	100+ trees average	Same but numbers
Neural Networks	Many layers	Most powerful

7. Unsupervised Learning

No labels. Computer finds patterns alone.

Clustering = group similar things

Anomaly Detection = find the weird one

8. Reinforcement Learning

Agent learns by trying + feedback. We define WHAT (goal). Model discovers HOW (strategy).

Term	Meaning	Example
Agent	Learner	You / Batsman
Environment	World	Road / Pitch
Reward	Good feedback	Balanced! / Four!
Penalty	Bad feedback	Fell / Got out

3 Types Comparison

Type	Answers	Cricket
Supervised	WHAT is this?	Ball, bat, bowler
Unsupervised	What PATTERNS?	Fielding positions
Reinforcement	HOW to do well?	Hit or defend?

Real Projects Use All 3

Exam Bot: Supervised (recognize cheating) + Unsupervised (spot unusual) + RL (when to alert)

Self-Driving: Priority system. Human life >> Injury >> Car damage. Evaluates ALL options in 0.001 seconds.

9. Neural Networks

Layer	What	Analogy
Input	Receives raw data	Reception
Hidden	Processing/learning	Departments
Output	Final answer	CEO

- Epoch = one complete pass through ALL training data
- Activation Function = bouncer at EVERY neuron deciding "pass forward or block?"

10. Math Foundations for ML

SPEAK (vectors) -> COMPARE (dot product) -> FIND (derivatives) -> FIX (gradient descent)

Math Tool	Job	When Used
Vector	Data -> numbers	Always first
Dot Product	Compare similarity	When comparing
Derivative	Find what is wrong	Every wrong guess
Gradient Descent	Fix weights step by step	Every training round

Train/Test Split

80% training, 20% testing. Computer checks accuracy automatically.

Overfitting

Memorized training data, fails on new data. 99% training + 30% test = Overfitting. Bad data = low on BOTH.

- ML engineers: 60% data quality, 30% training, 10% model structure

11. How Math Connects to ML

Real Estate Agent Analogy: Start dumb, learn from mistakes, become expert after 1000 houses.

5 Things to Remember

- Model starts DUMB (random weights)
- Each round: guess -> wrong -> figure out why -> adjust
- After thousands of rounds -> EXPERT
- Discovers rules NOBODY taught it (old house = cheaper)
- Test on unseen data -> if accurate -> deploy!

You = Driver (decide problem). Math = Engine (calculations). Python = Car (press buttons).

12. Problem Understanding & Model Selection

5 Questions Before Any ML Project

#	Question	ExamGuard Example
1	What is my goal?	Detect cheating through cameras
2	What data do I have?	Labeled clips + unlabeled feeds
3	Answer looks like?	Word = cheating/normal
4	How much data?	10K+ clips = DL territory
5	Real-time needed?	YES = need fast models

Decision Flowchart

- Labeled data + Answer is WORD -> Classification
- Labeled data + Answer is NUMBER -> Regression
- No labels + Find groups -> Clustering
- No labels + Spot weird -> Anomaly Detection
- Goal + Feedback -> Reinforcement Learning

Classification Model Selection

Situation	Best Model	Why
Small data, simple	Logistic Regression	Fast, explainable
Need to explain reasoning	Decision Tree	Step-by-step logic
Best accuracy	Random Forest	Many trees vote
Clear boundary	SVM	Best dividing line
Image/Video data	CNN	Only option for images
Text data	Transformer/LLM	Only option for language
Behavior over time	LSTM/RNN	Understands sequences

Regression Model Selection

Situation	Best Model	Why
Simple straight-line	Linear Regression	Fast, explainable
Curved relationship	Polynomial	Captures curves
Complex, many features	Random Forest	Handles complexity
Massive data	Neural Network	Most powerful

Unsupervised Model Selection

Situation	Best Model	Why
Group similar things	K-Means	Simple, fast
Unknown group count	DBSCAN	Auto-finds groups
Spot frauds/outliers	Isolation Forest	Great at finding odd ones
Learn normal pattern	Autoencoder	Compressed normal baseline

6 Expert Tricks

- **Trick 1:** Start SIMPLE, go complex only if needed
- **Trick 2:** Data size determines model: <1K=simple, 1K-100K=medium, 100K+=DL

- **Trick 3:** Check if someone solved it already (YOLO, BERT, etc.)
- **Trick 4:** Transfer Learning: use pre-trained models, fine-tune with YOUR data
- **Trick 5:** Map each sub-problem separately (ExamGuard has 7 sub-problems)
- **Trick 6:** Always test 3-4 models, pick best accuracy + speed

ExamGuard Model Map

Sub-Problem	ML Type	Model	Why
Detect phone	Supervised	YOLO	Pre-trained, real-time
Where looking?	Supervised	CNN	Best for images
Cheating behavior?	Supervised	CNN + LSTM	Frames + time sequence
Unusual behavior?	Unsupervised	Autoencoder	Learns normal, flags rest
Group behaviors	Unsupervised	K-Means	Simple, proven
When to alert?	RL	Custom RL	Balances alert vs ignore
Track across cameras	Deep Learning	Re-ID Network	Matches person from angles